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from google.colab import files

uploaded = files.upload()
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.backends.backend_pdf import PdfPages

sns.set_style("whitegrid") # optional: gives cleaner-looking charts
df = pd.read_csv("placement_predict_50k Dataset.csv")
print("First 5 rows of the dataset:")
display(df.head()) # use display() in Colab; use print(df.head()) elsewhere
print("\nShape of dataset (rows, columns):", df.shape)

print("\nColumn names:")
print(df.columns.tolist())

print("\nData types and non-null counts:")
df.info()

print("\nMissing values per column:")
print(df.isnull().sum())
df['Placed_Flag'] = df['PlacementStatus'].apply(lambda x: 1 if x == 'Placed' else 0)
num_cols = ['CGPA', 'AttendancePercent', 'AptitudeTestScore',
            'CodingTestScore', 'SoftSkillsRating', 'Salary Package']
fig1 = plt.figure(figsize=(10, 6))
sns.heatmap(df.isnull(), cbar=False, cmap="viridis", yticklabels=False)
plt.title("Missingness Heatmap - Placement Prediction Dataset")
plt.xlabel("Columns")
plt.tight_layout()
plt.show()
fig2, axes = plt.subplots(2, 3, figsize=(15, 8))

for i, col in enumerate(num_cols):
    row, colpos = divmod(i, 3)
    sns.histplot(df[col].dropna(), bins=30, kde=True,
                 ax=axes[row, colpos], color='steelblue')
    axes[row, colpos].set_title(f'Distribution of {col}')

plt.tight_layout()
plt.show()

fig3 = plt.figure(figsize=(10, 6))

placement_rate = (
    df.groupby('Stream')['Placed_Flag'].mean().sort_values(ascending=False) * 100
)

sns.barplot(x=placement_rate.index, y=placement_rate.values, palette='crest')
plt.title('Placement Rate (%) by Stream')
plt.xlabel('Stream')
plt.ylabel('Placement Rate (%)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

print("\nPlacement rate (%) by stream:")
print(placement_rate.round(2))
fig4 = plt.figure(figsize=(10, 8))

numeric_df = df.select_dtypes(include=['int64', 'float64'])
corr_matrix = numeric_df.corr()

sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm',
            square=True, linewidths=0.5)
plt.title('Correlation Heatmap - Numerical Features')
plt.tight_layout()
plt.show()
fig5, axes5 = plt.subplots(2, 3, figsize=(15, 8))

for i, col in enumerate(num_cols):
    row, colpos = divmod(i, 3)
    sns.boxplot(y=df[col], ax=axes5[row, colpos], color='lightcoral')
    axes5[row, colpos].set_title(f'Boxplot of {col}')

plt.tight_layout()
plt.show()
pdf = PdfPages('EDA_Report.pdf')

pdf.savefig(fig1) # Page 1: Missingness Heatmap

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pdf.savefig(fig1) # Page 1: Histograms
pdf.savefig(fig2) # Page 2: Six Histograms
pdf.savefig(fig3) # Page 3: Placement Rate by Stream
pdf.savefig(fig4) # Page 4: Correlation Heatmap
pdf.savefig(fig5) # Page 5: Boxplots

pdf.close()

print("\nEDA_Report.pdf created successfully with 5 pages!")
files.download('EDA_Report.pdf')
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